

Introducción al Prompt Engineering

Técnicas esenciales para trabajar con modelos de lenguaje de IA

UNGS - Matemática para Economistas III

Lic. Mateo Suster

A.I. TURNS THIS SINGLE
BULLET POINT INTO A
LONG EMAIL I CAN
PRETEND I WROTE.



A.I. MAKES A SINGLE
BULLET POINT OUT OF
THIS LONG EMAIL I CAN
PRETEND I READ.



TOM
FISH
BURNÉ

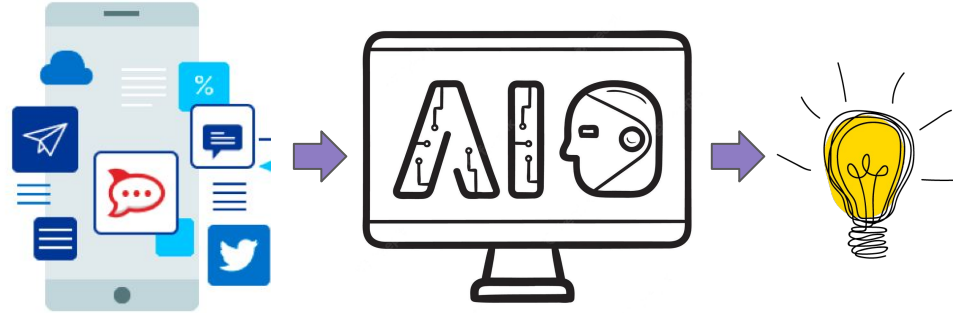
Agenda

- Natural Language Processing (NLP)
- Tareas de NLP
- How the models works?
- Prompt Engineering

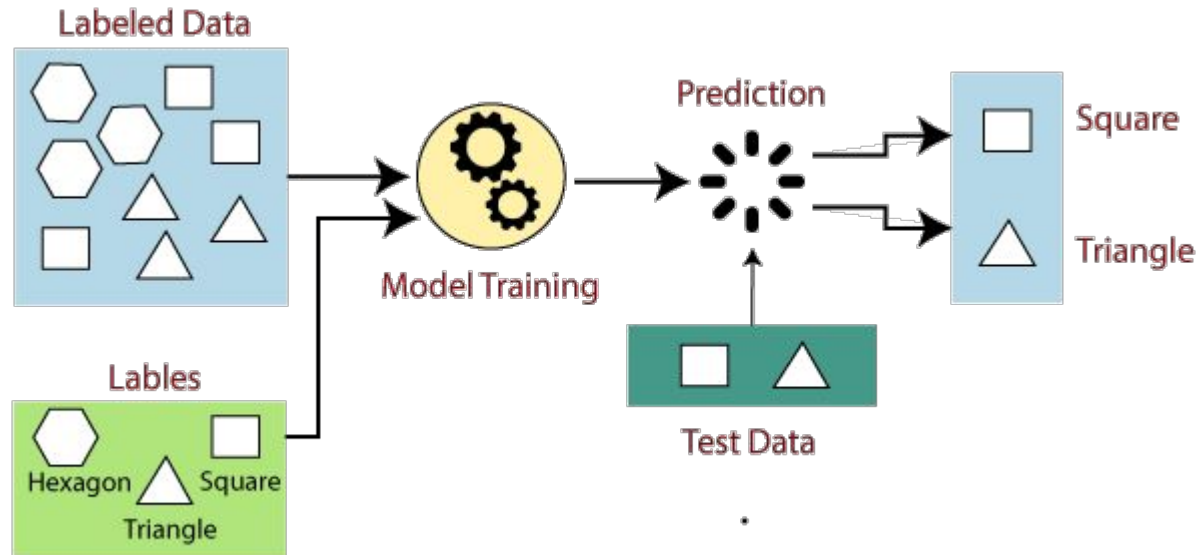
What is NLP?

Definition of NLP

- NLP stands for Natural Language Processing.
- NLP is a field of computer science that is focused on **enabling computers** to **understand** and **manipulate** natural language to do **useful things**.



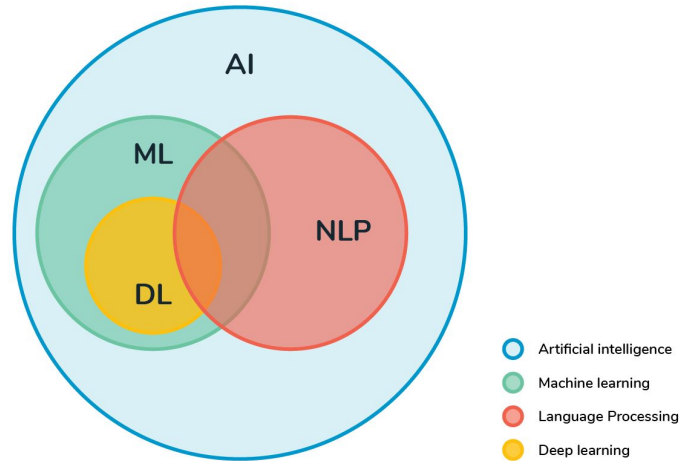
Machine Learning - Supervised learning



What is a model?



Relationship between NLP and Artificial intelligence



NLP use cases

Machine translation

Google Translate is an example of widely available NLP technology about machine translation. Effective translation has to capture accurately the meaning and tone of **the input language and translate it to text with the same meaning** and desired impact in the output language.



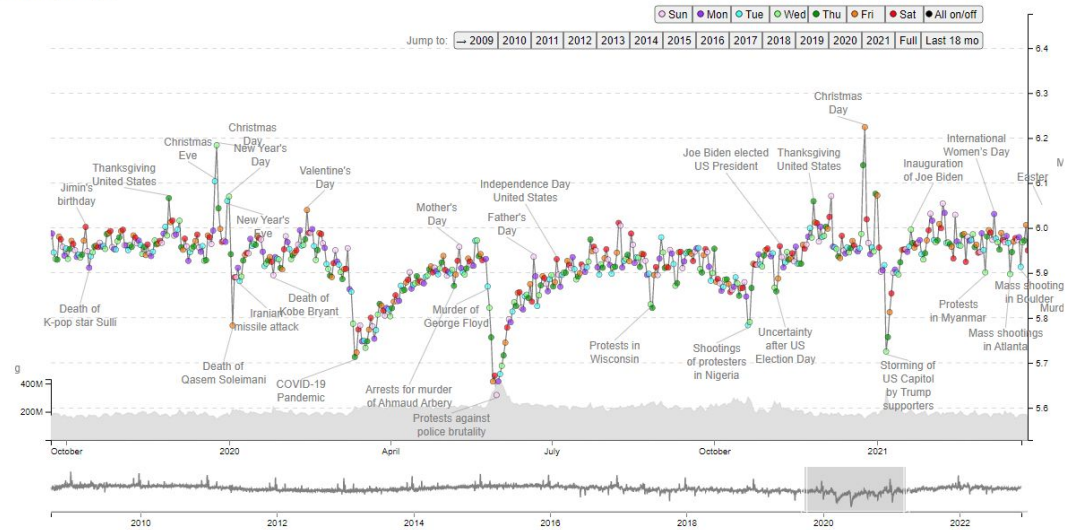
Sentiment analysis

Sentiment analysis can analyze language used in social media posts, responses, reviews, and more to **extract attitudes and emotions** in response to products, promotions, and events.

Information companies can use it in product designs, advertising campaigns, and more.

Average Happiness for Twitter

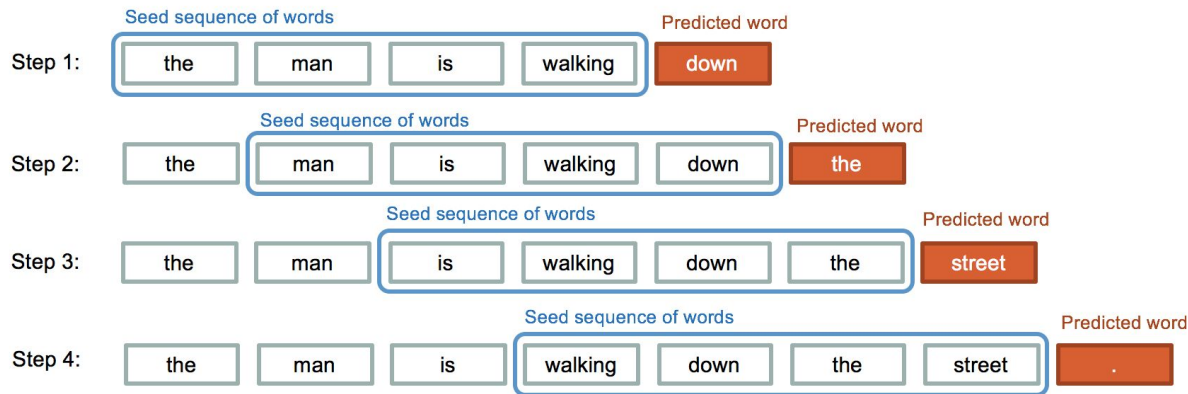
[All Tweets in English](#)



<https://hedonometer.org/>

Text generation

Text generation leverages knowledge in computational linguistics and artificial intelligence to **automatically generate natural language texts**, which can satisfy certain communicative requirements.



Other tasks...

- Text classification
- Information retrieval
- Question answering
- Summarization
- Automatic speech recognition
- Named Entity Recognition (NER)
- Word Sense Disambiguation
- Information Extraction
- ...

How the models works?

AI chat

The diagram illustrates a chat interface with a light gray background. It features a central white rounded rectangle containing a conversation. The conversation consists of four messages: a 'You' message asking about quantum computing, an 'AI' response, another 'You' message asking for another sentence, and a final 'AI' response. At the bottom of the chat window is a white input field and a yellow 'Send' button.

You
What's quantum computing?
Answer in one sentence.

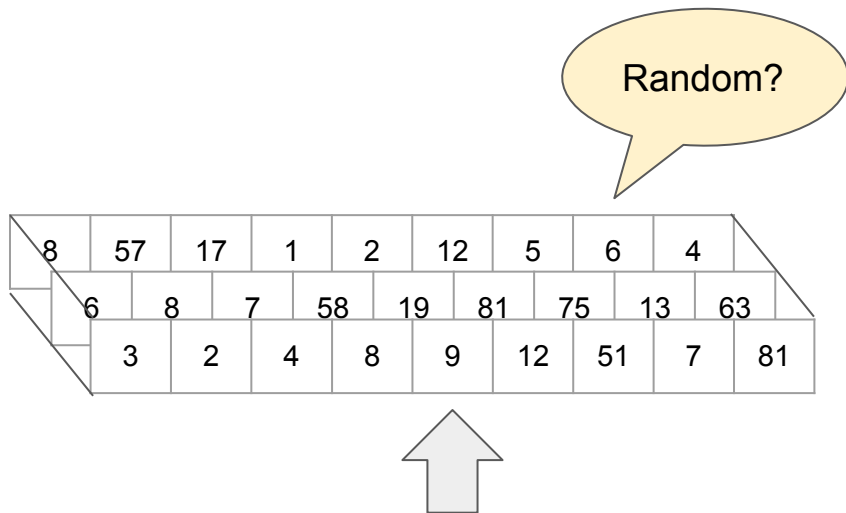
AI
Quantum computing is a type...

You
Write another sentence

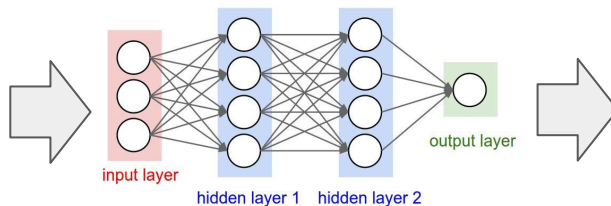
AI
The applications of quantum...

Send

What is an NLP model?



[The quick brown fox jumps over the lazy dog]

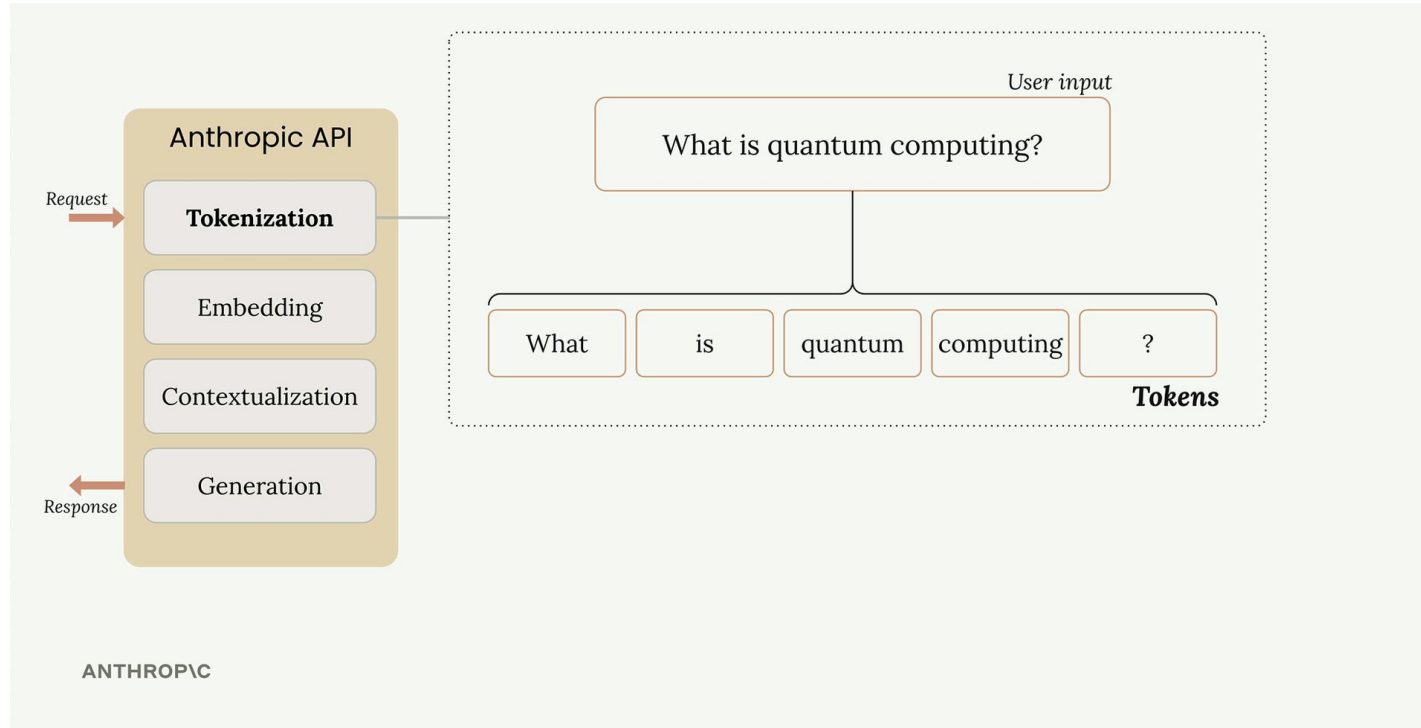


Model

Output:

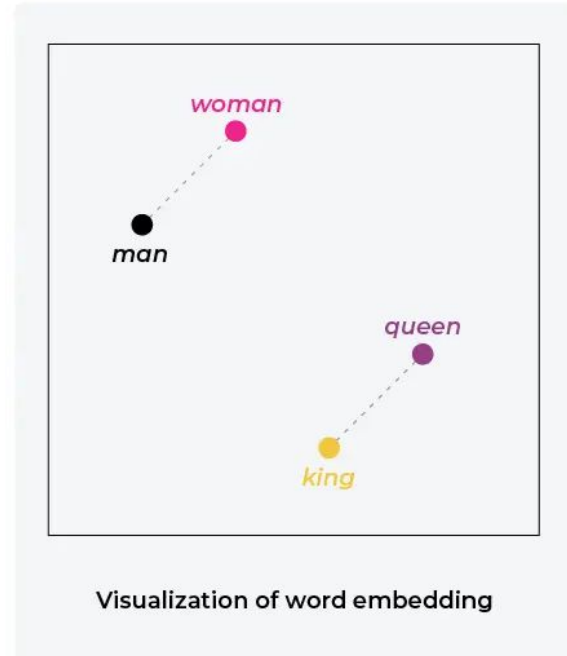
- Text classification
- Word prediction
- Translation
- ...

Tokenization

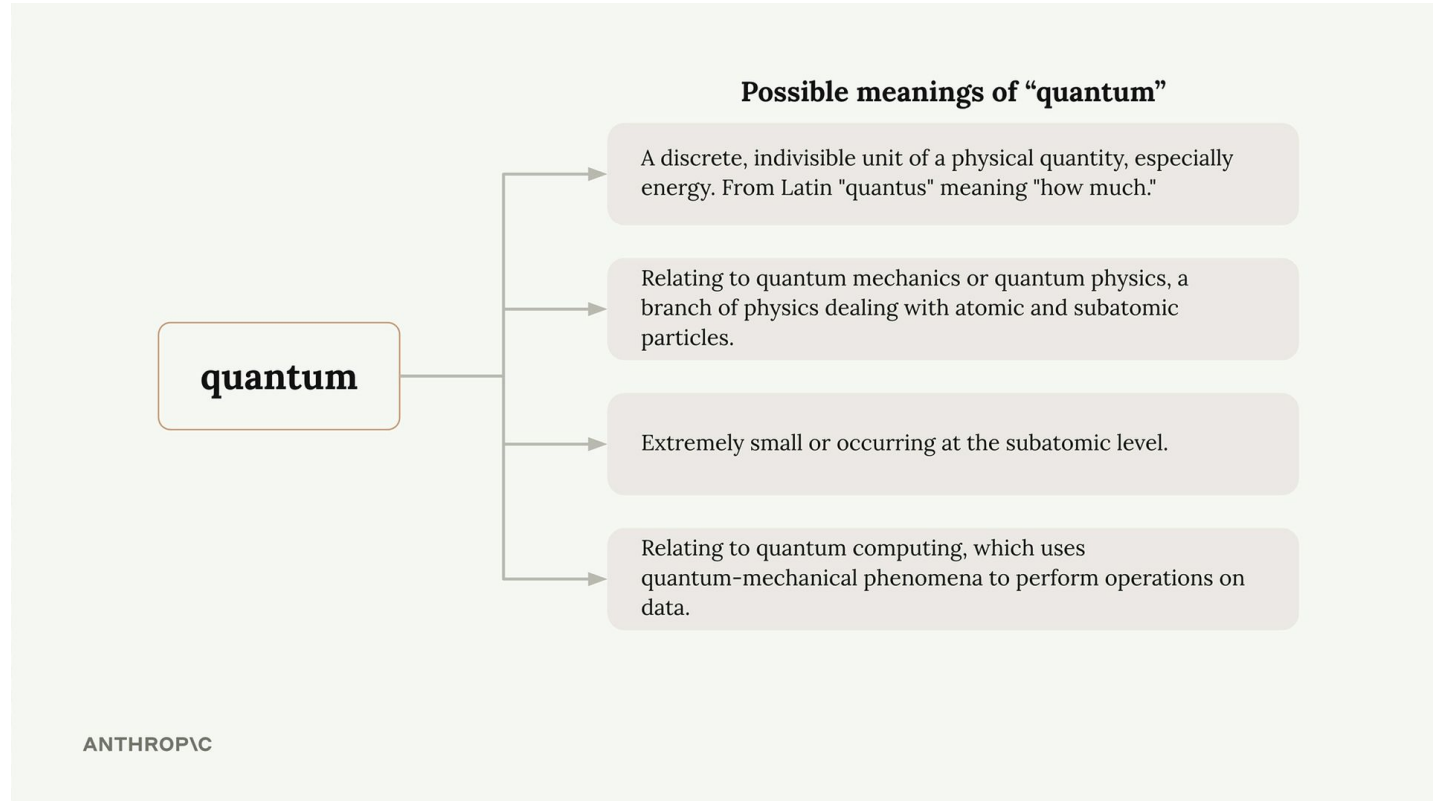


Word embeddings (I)

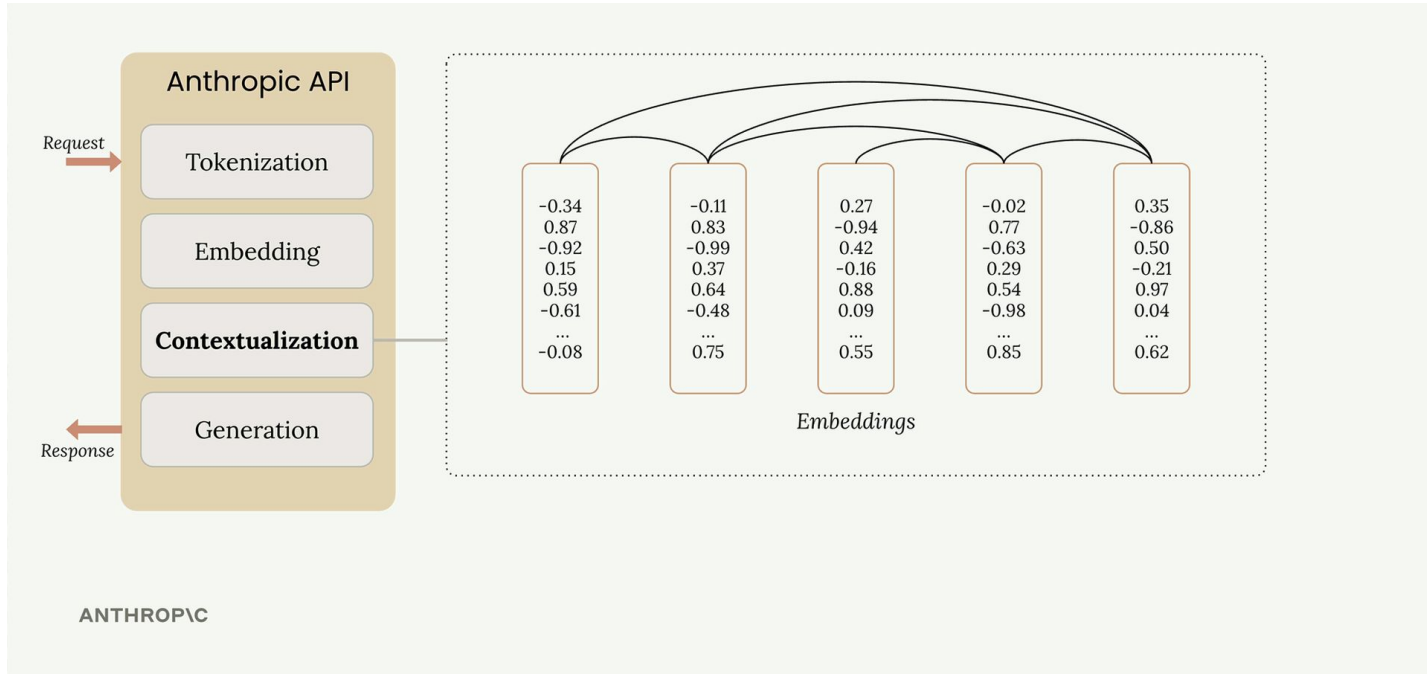
		living being	feline	human	gender	royalty	verb	plural
man	→	0.6	-0.2	0.8	0.9	-0.1	-0.9	-0.7
woman	→	0.7	0.3	0.8	-0.7	0.1	-0.5	-0.4
king	→	0.5	-0.4	0.7	0.8	0.9	-0.7	-0.6
queen	→	0.8	-0.1	0.8	-0.9	0.8	-0.5	-0.9
word		Word embedding						



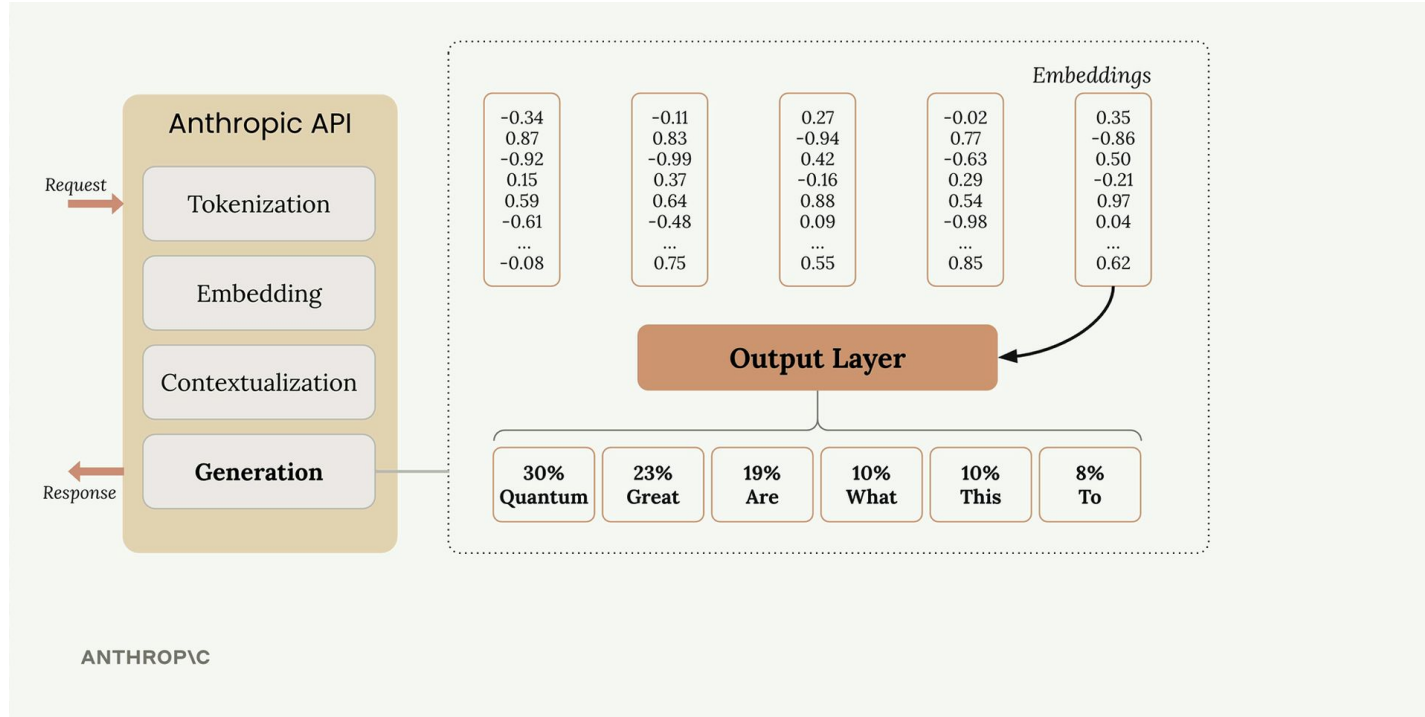
Word embeddings and contextualization



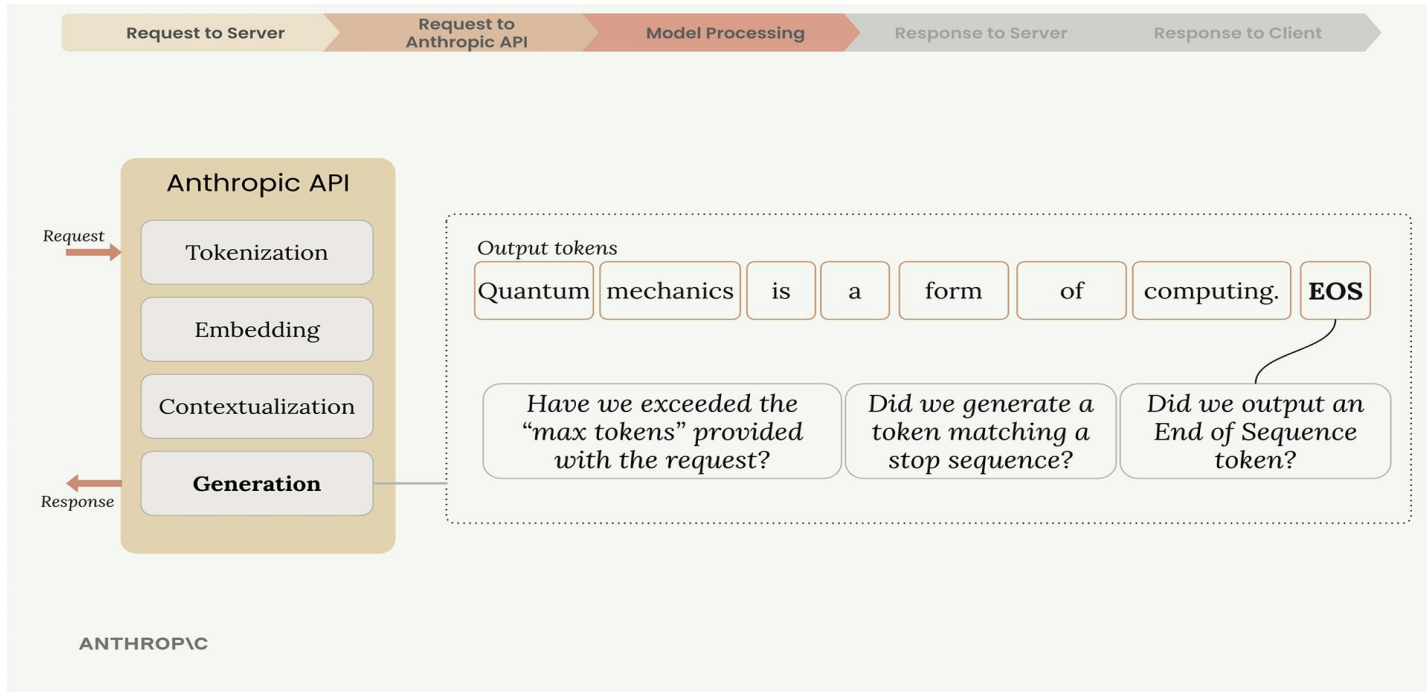
Contextualization



Generation (I)



Generation (II)



Prompt Engineering

¿Qué es el Prompt Engineering?

La **ingeniería de prompts** es una nueva disciplina para desarrollar y optimizar prompts para usar de manera eficiente los modelos de lenguaje (LM) en una amplia variedad de aplicaciones.

Las habilidades de **ingeniería de prompts** ayudan a comprender mejor las capacidades y limitaciones de los grandes modelos de lenguaje (LLM) y, por lo tanto, a lograr mejores resultados en nuestros objetivos.

El **objetivo final** es crear entradas (prompts) que guíen a los modelos de lenguaje hacia respuestas específicas y de alta calidad de los modelos de lenguaje, ayudando a mejorar la **precisión, relevancia y creatividad** de las respuestas del modelo.

Ejemplo de prompt

Here's an example of a prompt:



Prompt

 Copy

What is $923 * 99$?

Let's see how it works:



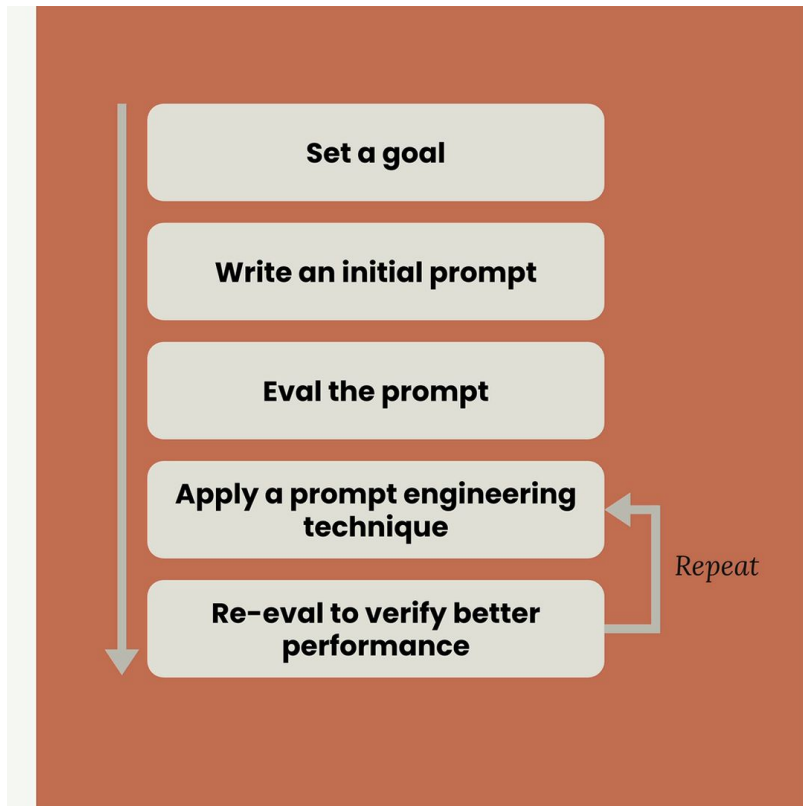
AI Output

 Copy

$923 * 99$ equals 91,677.

How we learn prompt engineering

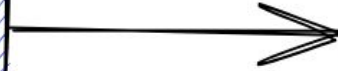
Write an initial prompt, then improve it **step by step** while learning new techniques.



Instruction prompting

An Instruction Prompt

Write the
word "funny"
backwards



Model Output

ynnuf

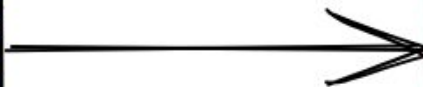
Role assignment

A Role Prompt

You are
Shakespeare, an
English writer.
Write me a poem.

Model Output

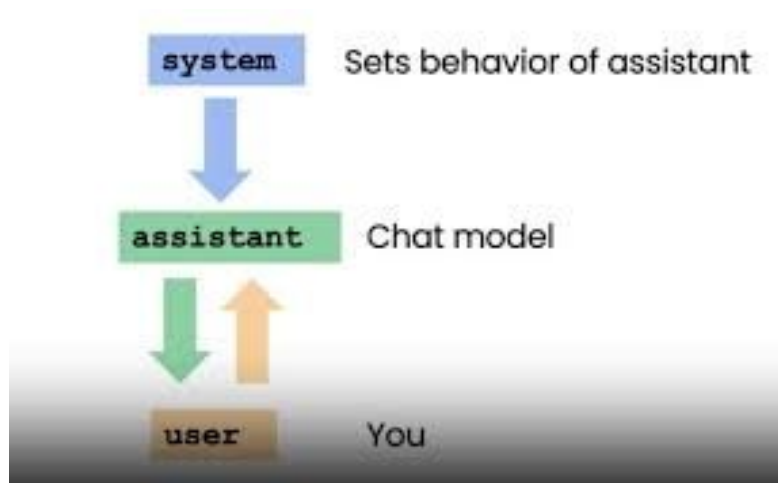
Of lovers' hearts
and passion's
fire...



Roles

Cuando usás LLMs, los prompts generalmente se componen de los siguientes elementos:

- **Sistema (System):** Este elemento guía y estructura el comportamiento de la IA, asegurando que responda de manera apropiada a los prompts del usuario. Define el contexto, el tono y los límites de la interacción.
- **Usuario (User):** Representa a la persona que interactúa con el modelo de IA.
- **Asistente (Assistant):** Es el contenido generado por la IA que sirve como respuesta a la entrada del usuario.



Estrategias para un prompt efectivo

Be clear and direct

The first line of your prompt is the most important part of your entire request...

"Clear"

- Use simple language
- State what you want explicitly
- Lead your prompt with a simple statement of the model's task

Instead of: "I need to know about those things people put on their roofs that use sun - those solar panel things, I think they're called"

Use: "Write three paragraphs about how solar panels work"

"Direct"

- Use instructions, not questions
- Use direct action verbs ("Write", "Create", "Generate")

Instead of: "I was reading about renewable energy and geothermal energy sounds neat. What countries use it?"

Use: "Identify three countries that use geothermal energy. Include generation stats for each."

Escribe instrucciones claras

- Incluye detalles en tu consulta para obtener respuestas más relevantes.
- Pídele al modelo que adopte una personalidad.
- Usa delimitadores para indicar claramente las distintas partes de la entrada.
- Especifica los pasos necesarios para completar una tarea.
- Proporciona ejemplos.
- Especifica la longitud deseada de la salida.

Be specific

Provide a list of guidelines or steps to direct the model

Not Great

```
prompt = f"""  
Write a short story about a  
character who discovers a hidden  
talent  
"""
```



Better!

```
prompt = f"""  
Write a short story about a character who  
discovers a hidden talent  
  
Guidelines:  
1. Keep the story under 1,000 words  
2. Include a clear action that reveals the  
   character's talent  
3. Include at least one supporting  
   character  
"""
```

ANTHROPIC

Two Types of Guidelines

List qualities that the output should have

Write a short story about a character who discovers a hidden talent

Guidelines:

1. Keep the story under 1,000 words
2. Include a clear action that reveals the character's talent
3. Include at least one supporting character

Provide steps the model should follow

Write a short story about a character who discovers a hidden talent

Follow these steps:

1. Brainstorm 3 talents that would create dramatic tension
2. Pick the most interesting talent
3. Outline a pivotal scene that reveals the talent
4. Brainstorm 3 supporting characters types that could increase the impact of this discovery

When to Use Each Approach

Always Use Output

Guidelines

You should include quality guidelines **in almost every prompt** you write. They're your safety net for getting consistent, useful results.

Use Process Steps For

Complex Problems

Add **step-by-step instructions** when you're dealing with:

- Troubleshooting complex problems
- Decision-making scenarios
- Critical thinking tasks
- Any situation where you want Claude to consider multiple angles

Combining Both

Approaches

In professional prompting, you'll often see **both techniques used together**.

Structure with XML tags

Use XML tags to separate distinct portions of the prompt.

XML tags are most useful when:

- Including **large amounts of context** or data
- **Mixing different types of content** (code, documentation, data)
- You want to be **extra clear** about content boundaries
- Working with **complex prompts** that interpolate multiple variables

Even for shorter content, XML tags can help serve as delimiters that make your prompt structure more obvious to the model.

```
Write a one page decision report to troubleshoot  
why a sales team's numbers have dropped 30% last  
quarter.
```

```
Here are the last 20 pages of our sales records:
```

```
<sales_records>  
{sales_records}  
</sales_records>
```

```
Follow these steps:
```

1. Compare current vs previous market metrics
2. Identify relevant industry changes
3. Analyze individual team member performance
4. Consider recent organizational changes
5. Review customer feedback

Providing structure

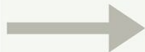
Not Great

```
Debug my code below using the provided
documentation.

from datavortex import Pipeline, DataSource

def process_data(input_file, output_file):
    pipeline = Pipeline()
    source = DataSource.from_csv(input_file)

# Creating a data source from data vortex
csv_source = DataSource.from_csv("data.csv")
```



Better!

Debug my code below using the provided documentation.

```
<my_code>
from datavortex import Pipeline, DataSource

def process_data(input_file, output_file):
    pipeline = Pipeline()
    source = DataSource.from_csv(input_file)
</my_code>

<docs>
# Creating a data source from data vortex
csv_source = DataSource.from_csv("data.csv")
</docs>
```

Custom Tag Names

You don't need to use official XML tags. Create descriptive names that make sense for your content:

- `<sales_records>` is better than `<data>`
- `<athlete_information>` clearly identifies user details
- `<my_code>` and `<docs>` separate different types of content

The more specific and descriptive your tag names, the better the model can understand the purpose of each section.

Divide tareas complejas en subtareas más simples

- Utiliza la clasificación de intenciones para identificar las instrucciones más relevantes para una consulta del usuario.
- Para aplicaciones de diálogo que requieren conversaciones muy largas, resume o filtra el diálogo anterior.
- Resume documentos largos por partes y construye un resumen completo de forma recursiva.

Dale tiempo al modelo para "pensar"

- Instruye al modelo a elaborar su propia solución antes de apresurarse a una conclusión.
- Usa el monólogo interno o una secuencia de consultas para ocultar el proceso de razonamiento del modelo.
- Pregúntale al modelo si se perdió algo en pasadas anteriores.

Usa herramientas o fuentes externas

- Instruye al modelo a responder utilizando citas o texto de referencia.
- Utiliza la búsqueda basada en embeddings para implementar una recuperación de conocimiento eficiente.
- Utiliza la ejecución de código para realizar cálculos más precisos o llamar a API externas.
- Dale al modelo acceso a funciones específicas.

Providing examples

The challenge here is sarcasm. A tweet like *"Yeah, sure, that was the best movie I've seen since 'Plan 9 from Outer Space'"* appears positive on the surface, **but it's actually sarcastic and negative** (Plan 9 is famously one of the worst movies ever made).

Categorize the sentiment of the below tweet:

```
<input_tweet>
```

```
Yeah, sure, that was the best movie I've seen  
since 'Plan 9 from Outer Space'
```

```
</input_tweet>
```

If the tweet has a positive sentiment, respond with "Positive". If it is negative, respond with "Negative".

Adding Examples to Handle Corner Cases

Input Containing Sarcasm

Categorize the sentiment of the below tweet:

```
<input_tweet>
Yeah, sure, that was the best movie I've seen
since 'Plan 9 from Outer Space'
</input_tweet>
```

If the tweet has a positive sentiment,
respond with "Positive". If it is negative,
respond with "Negative".

ANTHROPIC

Categorize the sentiment of the below tweet:

```
<input_tweet>
Yeah, sure, that was the best movie I've seen since 'Plan 9 from Outer
Space'
</input_tweet>
```

If the tweet has a positive sentiment, respond with "Positive". If it is
negative, respond with "Negative".

Here is a example input with an ideal response:

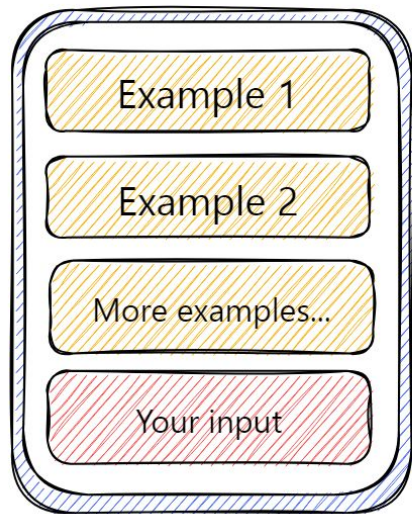
```
<sample_input>
Great game tonight!
</sample_input>
<ideal_output>
Positive
</ideal_output>
```

Be especially careful with tweets that contain sarcasm.
For example:

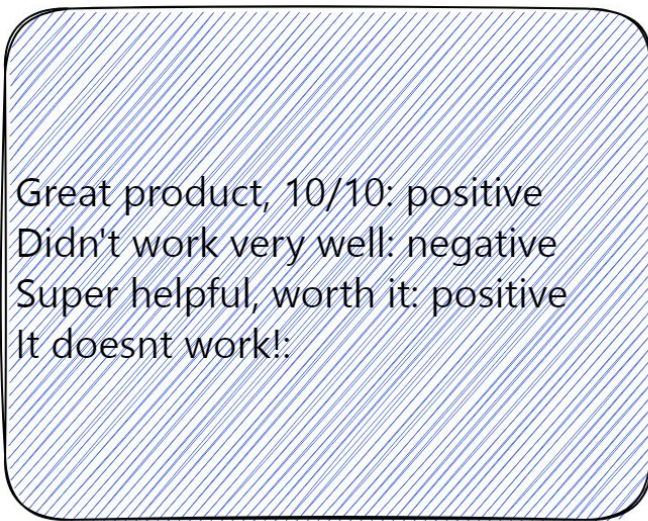
```
<sample_input>
Oh yeah, I really needed a flight delay tonight! Excellent!
</sample_input>
<ideal_output>
Negative
</ideal_output>
```

Few Shot-Based Prompting

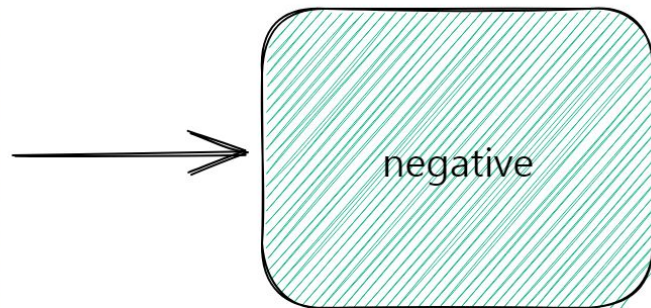
A Few Shot Prompt



Example



Model Output



What is Zero-Shot Prompting?



Zero-Shot Prompt



Classify the sentiment of the following text as positive, negative, or neutral.

Text: I think the vacation was okay.

Sentiment:

The model will provide a classification based solely on the task description, without seeing any examples beforehand. The output might be:



AI Output



Neutral

One-Shot Prompting



One-Shot Prompt



Copy

Classify the sentiment of the following text as positive, negative, or neutral.

Text: The product is terrible.

Sentiment: Negative

Text: I think the vacation was okay. Sentiment:

Here, the model is shown a single example ("The product is terrible. Sentiment: Negative") before it processes the new input. This allows the model to better understand what it should do next. The output might now be more reliable:



AI Output



Copy

Neutral

Few-Shot Prompting



Few-Shot Prompt



Classify the sentiment of the following text as positive, negative, or neutral.

Text: The product is terrible. Sentiment: Negative

Text: Super helpful, worth it Sentiment: Positive

Text: It doesnt work! Sentiment:

Here is the output of this prompt when passed through GPT-4. In this case, the model has two examples to learn from, making it more likely to generate an accurate response for the new input:



AI Output



Negative

Chain-of-Thought (CoT) Prompting

Técnica que mejora las capacidades de razonamiento de los grandes modelos de lenguaje (LLM). Lo logra al incorporar pasos lógicos —o una "cadena de pensamiento"— dentro de la indicación misma.

Standard Prompting

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain of Thought Prompting

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

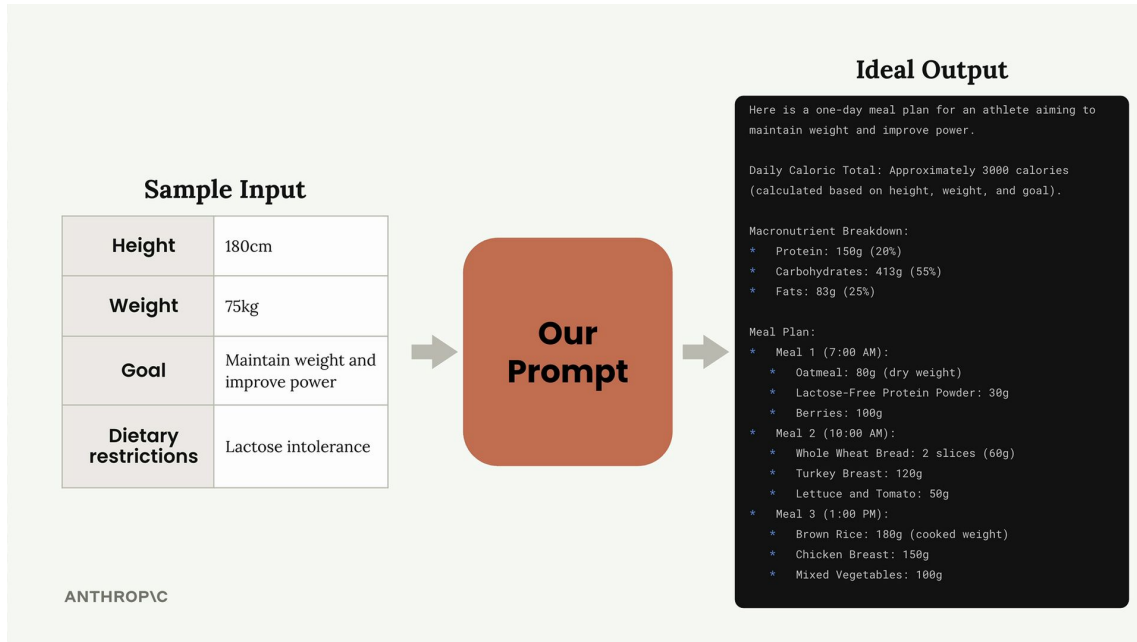
Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Prueba los cambios sistemáticamente

- Evalúa las salidas del modelo con referencia a respuestas estándar de oro.



Síntesis de un buen prompt

Un buen "prompt" para un usuario contiene los siguientes elementos:

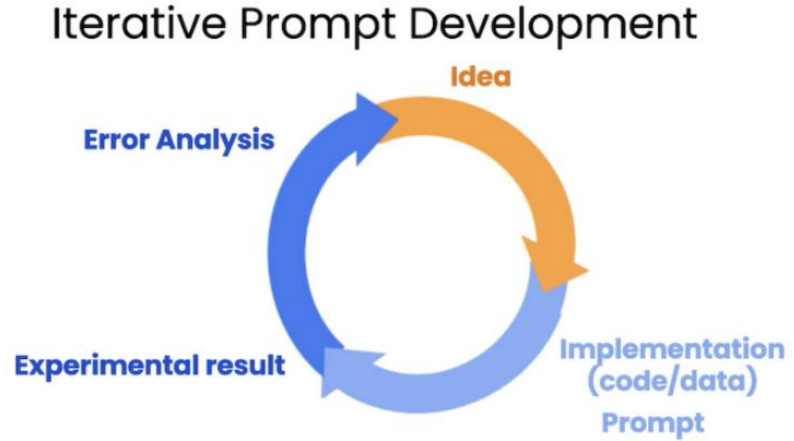
- **Instrucción:** Una tarea o instrucción específica que quieres que el modelo realice.
- **Contexto:** Información externa o contexto adicional que puede guiar al modelo para obtener mejores respuestas.
- **Datos de entrada:** La entrada o pregunta para la que estamos interesados en encontrar una respuesta.
- **Indicador de salida:** El tipo o formato de la respuesta esperada.

Errores Comunes en la Ingeniería de Prompts

- **Prompts Vagos o Ambiguos:**
 - Evita usar frases poco claras o abiertas.
- **Prompts Demasiado Complejos:**
 - Mantén los prompts simples y enfocados en la tarea principal.
- **Falta de Instrucciones Específicas:**
 - Proporciona pautas claras sobre formato, estilo y tono.
- **Ignorar las Limitaciones del Modelo:**
 - Sé consciente de las capacidades y limitaciones del modelo.
- **No realizar iteraciones o refinaciones frente a resultados no deseados**
 - Experimenta con diferentes prompts y ajusta según sea necesario.

Prompt Engineering es un **proceso iterativo**.

Casi nunca se acierta el prompt a la primera.....



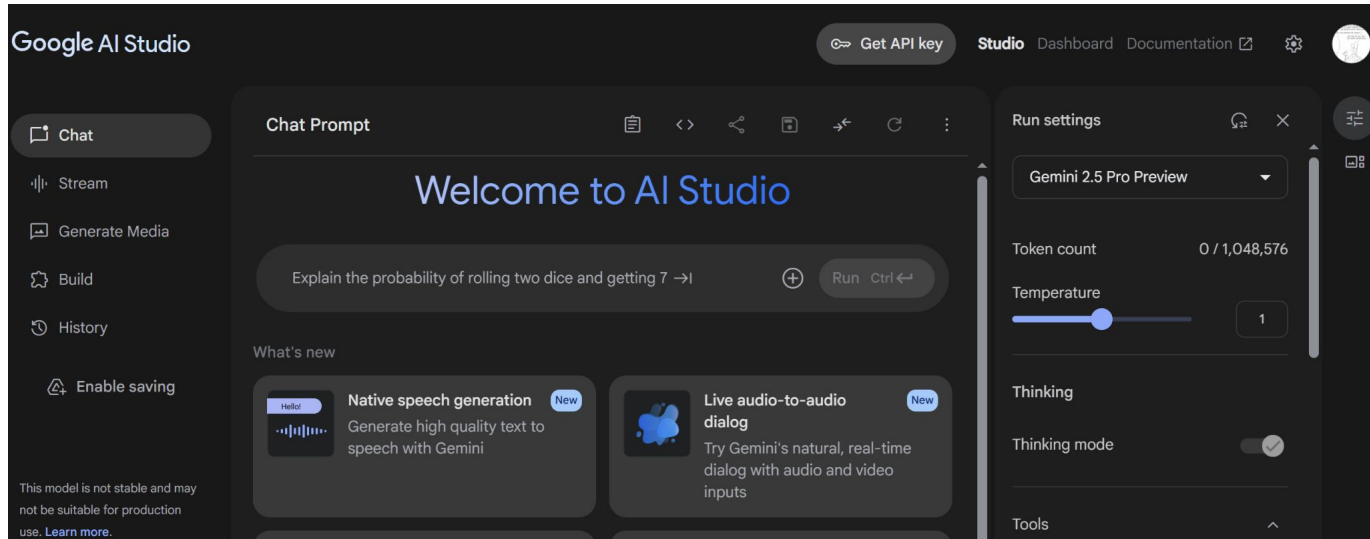
Iterative Process

- Try something
- Analyze where the result does not give what you want
- Clarify instructions, give more time to think
- Refine prompts with a batch of examples

Playground

Playground y configuraciones extras

Para personalizar completamente las capacidades de un LLM, no siempre es necesario desarrollar un script para acceder a todas las funcionalidades de la API. Algunos chatbots, como ChatGPT o [Google AI Studio](#), ofrecen un entorno de pruebas (o "playground") que permite a los usuarios configurar diversas características directamente.



Temperatura, Top-P y Longitud Máxima

Temperatura (Temperature): La temperatura regula la imprevisibilidad de la salida de un modelo de lenguaje. Valores más bajos (por ejemplo, $T = 0.1$) producen respuestas enfocadas y predecibles; mientras que valores más altos fomentan la aleatoriedad y la creatividad.

Top-P: Limita la salida a los tokens más probables dentro de una probabilidad acumulativa (por ejemplo, 0.9). Valores más bajos producen resultados más enfocados, mientras que valores más altos permiten una mayor diversidad. Puede combinarse con la temperatura para un control más preciso sobre la aleatoriedad y la creatividad.

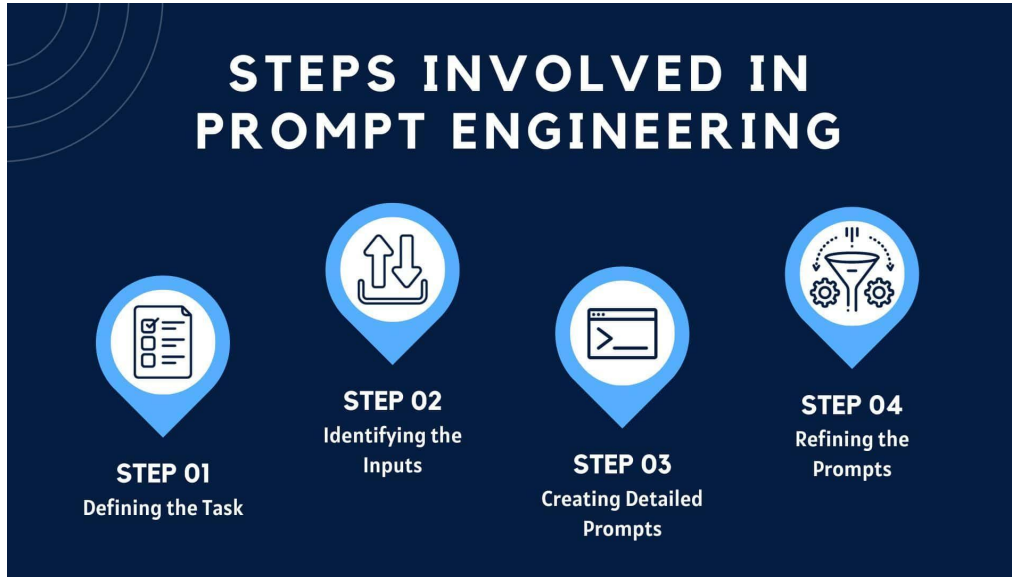
Longitud Máxima (Max Length): La longitud máxima es el número total de tokens que el LLM tiene permitido generar.

Recursos

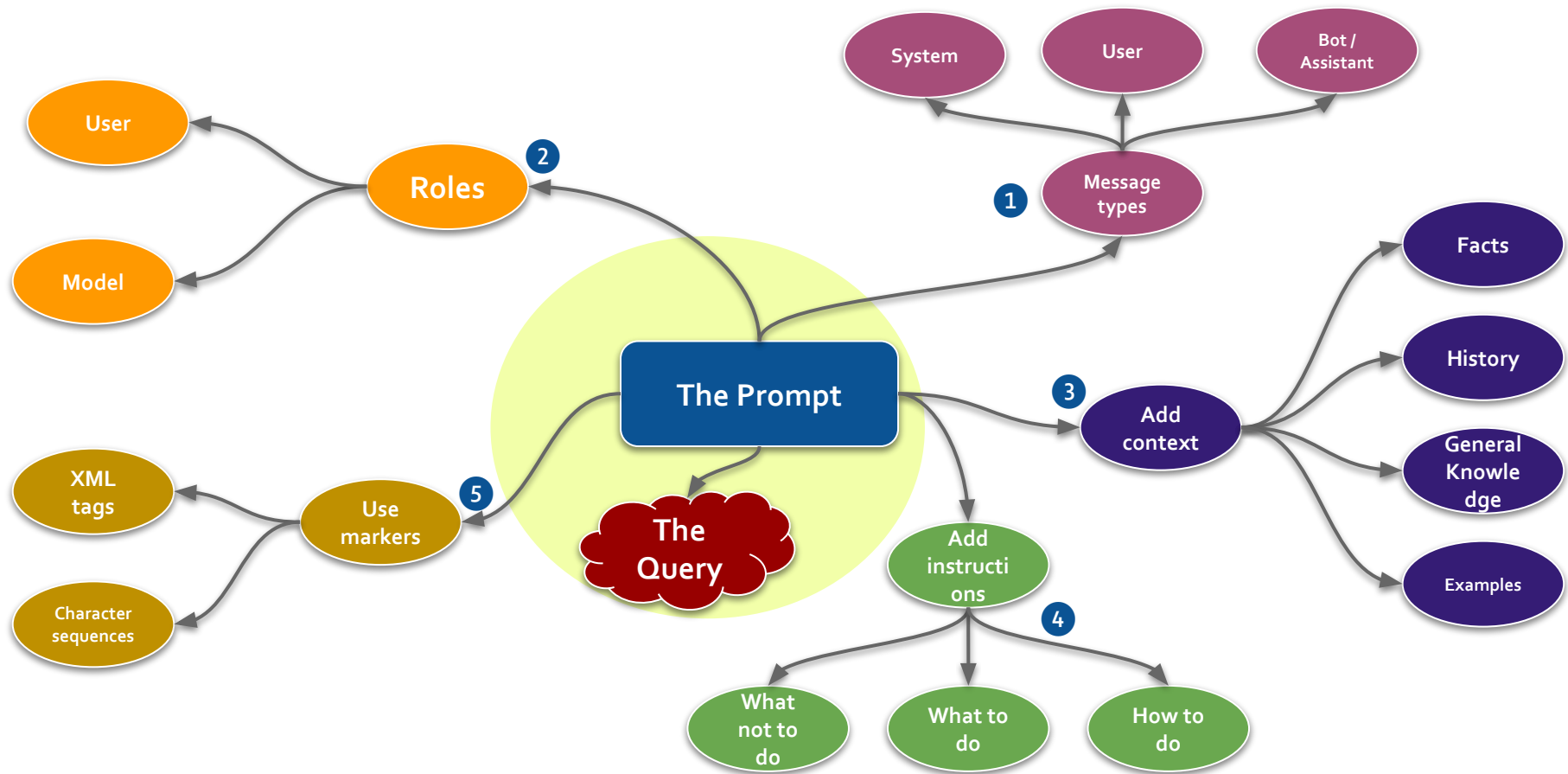
- **OpenAI prompting guide:** <https://platform.openai.com/docs/guides/prompt-engineering>
- **Guia básica de Prompt Engineering:** <https://www.promptingguide.ai/es>
- **Prompt Engineering Guide:** <https://learnprompting.org/docs/intro> (recomiendo el [siguiente post](#))
- **Prompting intro de Gemini:** <https://ai.google.dev/gemini-api/docs/prompting-intro>
- **Algunos chats con modelos de IA:**
 - [ChatGPT](#)
 - [Gemini](#) o [Google AI Studio](#)
 - [Claude](#)
 - [HuggingFace](#)

Appendix

Pasos en Prompt Engineering



Parts of the prompt



Elementos de un buen prompt

Actúa como un tutor que desglosa temas complejos en explicaciones fáciles.

Quiero que expliques el proceso de la fotosíntesis a un estudiante de 14 años de edad, para ayudarlo con la preparación del examen de biología.

Tu respuesta debe ser de 300 palabras, escrita en un tono amigable y educativo.

Rol: Pídele que asuma un rol

Objetivo: ¿Qué quieres que haga la IA?

Audiencia: Especifica a quién va dirigido

Contexto: ¿Qué necesita saber la herramienta?

Límites: Establece tu propia dirección y limitación

Consejo 1

Dar instrucciones claras

Utiliza comandos que indiquen a la herramienta de IA lo que desea generar, como "explicar", "traducir", "resumir" o "comparar".

Consejo 2

Proporcionar contexto

Agrega contexto e información que pueda ayudar a la herramienta a comprender mejor la tarea. Por ejemplo, menciona el tipo de proyecto, como "relato corto", "informe" o "esquema".

Consejo 3

Iterar y experimentar

Prueba diferentes instrucciones y técnicas si no obtienes los resultados que deseas. La indicación (*prompt*) puede ser como un experimento que puede requerir varias rondas de iteraciones.

GPT Example Prompt

01	Context	<u>Ignore the previous prompts in this conversation. You are an experienced content writer with high levels of expertise and authority within the tech industry. Your task is to write content that will be published online on websites, social media, email newsletters, and in advertisements. Your writing style is informative, friendly and engaging while incorporating humor and real-life examples. I will provide you with a topic or series of topics and you will come up with an engaging article outline for this topic. Do you understand?</u> <u>Rewrite using more natural, expressive language and include some examples to accompany this information.</u>
02	Task	
03	Instruction	
04	Clarify	
05	Refine	

ChatGPT Cheat Sheet

The ChatGPT Cheat Sheet



Act as a [ROLE]

1. Analyst
2. Teacher
3. Marketer
4. Advertiser
5. Mindset Coach
6. Therapist
7. Journalist
8. Inventor
9. Lawyer
10. Ghostwriter
11. Website Designer
12. Best Selling Author
13. Chief Financial Officer
14. Expert Copywriter
15. Prompt Engineer
16. Accountant
17. Project Manager
18. Sports Coach
19. Financial Analyst
20. Full Stack Developer
21. Linux Terminal
22. "Position" Interviewer
23. Excel Sheet
24. Plagiarism Checker
25. Relationship Coach
26. Recruiter
27. JavaScript Console

Create a [TASK]

1. Headline
2. Article
3. Essay
4. Book Outline
5. Email Sequence
6. Social Media Post
7. Product Description
8. Cover Letter
9. Blog Post
10. SEO Keywords
11. Summary
12. Video Script
13. Recipe
14. Sales Copy
15. Analysis
16. Ad Copy
17. Web Page

Show as [FORMAT]

1. A Table
2. A List
3. Summary
4. HTML
5. Code
6. Spreadsheet
7. Graphs
8. CSV file
9. Plain Text file
10. JSON
11. Rich Text
12. PDF
13. XML
14. Markdown
15. Gantt Chart
16. Word Cloud
17. Emojis
18. An Analogy
19. Bullet Points

Set restrictions

1. Use poetic language
2. Adopt a formal tone
3. Write short sentences
4. Code only in HTML/CSS
5. Use Shakespearian style
6. Write using basic English
7. Use only scientific sources
8. Add pop culture references

Prompts For Biz Owners

1. Give me a list of inexpensive ideas on how to promote my business better.
2. Acting as a Business Consultant, What is the best way to solve this problem of [Problem]?
3. Create a 30-Day Social Media Content Strategy based on [Topic 1] & [Topic 2].

Prompts for Developers

1. Develop an architecture and code for a <description> website with JavaScript.
2. Help me find mistakes in the following code <paste code below>
3. I want to implement a sticky header on my website. Can you provide an example using CSS and JavaScript?
4. Please continue writing this code for JavaScript <past code below>

Prompts for Marketers

1. Can you provide me with some ideas for blog posts about [topic]?
2. Write a product description for my [product or service or company].
3. Suggest inexpensive ways I can promote my [company] without using social media.
4. How can I obtain high-quality backlinks to improve the SEO of [website name]?

Prompt Priming

1. ZERO - "Write me 5 Headlines about [Topic]"
2. SINGLE - "Write me 5 Headlines about [Topic]. Here is an example of one headline: 5 Ways to Lose Weight"
3. MULTIPLE - "Write me 5 Headlines about [Topic]. Here is an example of some headlines: 5 Ways to Lose Weight, How to Lose More Fat in 4 Weeks, Say Goodbye to Stubborn Fat, Find a faster way to Lose Weight Fast"

Prompts for Designers

1. Generate examples of UI design requirements for a [mobile app].
2. How can I design a [law firm website] in a way that conveys [trust and authority]?
3. What are some micro-interactions to consider when designing a Fintech app?
4. Create a text-based Excel sheet to input your copy suggestions. Assume you have 3 members in your UX writing team.

C.R.E.A.T.E Formula for Prompt Engineering

1. **C: Character** - Define the AI's role. E.g., "You are a seasoned copywriter with 20 years of experience in successful sales copy."
2. **R: Request** - Be specific with what you need. Instead of "Write a sales email for a sports car," specify, "Write a compelling email for the Cougar Hyper Sport, an electric car with top-tier acceleration."
3. **E: Examples** - Optionally, provide samples for more precise results. Headlines or tone examples can guide the AI's style.
4. **A: Adjustments** - Refine the prompt if it's not perfect. Use instructions like, "Avoid bullet points; use subheads."
5. **T: Type of Output** - Describe the desired format. E.g., "Produce a 500-word article with a title and conclusion."
6. **E: Extras** - Incorporate unique instructions.
 - Ex: "Ignore previous conversations."
 - Ex: "Ask questions before answering."
 - Ex: "Explain your thinking."
 - Ex: "Summarize what you know about me in the first person."
 - Ex: "Use data up to 2023."
 - Ex: "Only use reliable sources and cite them."
 - Ex: "Use CAPS LOCK for emphasis."

Using the C.R.E.A.T.E formula ensures better results, although it may require more time for prompt creation.

The ChatGPT Cheat Sheet

